

Test Case Coverage Verification — KAN-3: Create test cases for login functionality and SRS document

Status: DRAFT — pending human review **Source ticket:** [KAN-3](#) (AI_QA project) **Verified against:** the actual KAN-3 ticket fetched from JIRA (2026-08-17) and `output/KAN-3_Test_Cases.md` **Method:** 03-check-test-case skill — requirement-to-test-case traceability

1. Requirement Information

Field	Details
Project	AI_QA
Ticket	KAN-3 — Create test cases for login functionality and SRS document
Ticket Type	Task
Priority	Medium
Status	To Do
Feature / Module	Login functionality + SRS document verification
Reviewed Date	2026-08-17
Reviewed By	Test Case Coverage Verifier (skill)

2. Coverage Summary

Metric	Result
Requirements / Acceptance Criteria	3 (R1, R2, R3)
Applicable Coverage Points	3
Jira Test Cases Reviewed	32 (24 login + 8 SRS)
Fully Covered	3
Partially Covered	0
Uncovered	0
Ambiguous	0
Coverage	100%

Note: Coverage is computed at requirement level. The ticket contains **no explicit acceptance criteria** — coverage is derived from the ticket description only.

3. Requirement-to-Test Traceability

#	Requirement / Acceptance Criterion	Jira Test Case(s)	Coverage
1	R1 — Secure authentication: registered users securely authenticate before accessing authenticated features	TC-02, TC-04, TC-05, TC-06, TC-17, TC-18, TC-19, TC-20, TC-21, TC-24	Covered
2	R2 — Post-login access: product browsing, product customization, customer information, orders, order history, communication, payment-related functionality, delivery information	TC-03	Covered
3	R3 — Login page elements: email/mobile/username field, password field, Login button, Forgot Password, password visibility control, Remember Me	TC-01, TC-07, TC-08, TC-09, TC-10, TC-11, TC-12, TC-13, TC-14, TC-15, TC-16, TC-22, TC-23	Covered

4. Test Scenario Coverage

Scenario Type	Covered?	Test Case ID(s)
Positive scenarios	Yes	TC-02, TC-03
Negative scenarios	Yes	TC-04, TC-05, TC-06, TC-20, TC-21, TC-24
Mandatory field validation	Yes	TC-07, TC-08, TC-09
Optional field validation	Yes	TC-16 (Remember Me)
Boundary conditions	Yes	TC-10, TC-11, TC-12
Invalid data	Yes	TC-04, TC-05, TC-06, TC-22
Error handling	Yes	TC-17, TC-18
Authentication	Yes	TC-02, TC-19
Authorization	Yes	TC-24
Integration scenarios	NA	—
Business rules	NA	Ticket specifies no business rules
Data validation	Yes	TC-07, TC-08, TC-09, TC-10, TC-11, TC-12
Regression impact	NA	—
SRS document verification	Yes	SRS-01 to SRS-08

5. Covered Areas

- **R1 — Secure authentication:** fully covered by 10 test cases spanning valid login (TC-02), invalid identifier/password/both (TC-04/05/06), deterministic error messages (TC-17), no sensitive data exposure (TC-18), session creation (TC-19), unregistered account (TC-20), disabled/locked account (TC-21), and unauthenticated access blocking (TC-24).
- **R2 — Post-login access:** covered by TC-03, which walks all 8 authenticated functions listed in the ticket (product browsing, customization, customer info, orders, order history, communication, payment, delivery).
- **R3 — Login page elements:** covered by TC-01 (page loads with identifier field, password field, Login button) plus dedicated cases for password visibility (TC-14), Forgot Password (TC-15), Remember Me (TC-16), and field-level validation for empty/whitespace/single-char/max-length/special-char inputs (TC-07–12, TC-22) and case sensitivity (TC-13).
- **SRS deliverable:** 8 verification cases (SRS-01–08) check SRS existence, login coverage, acceptance-criteria testability, boundary/error-state coverage, non-functional requirements, terminology consistency, traceability, and versioning/change history.

6. Coverage Gaps

No requirement-level gaps were found. However, the following items are **blocking execution** (they are gaps in the ticket/spec, not in test coverage):

1. **No acceptance criteria** exist in the KAN-3 ticket — the requirements are derived from the description only.
 2. **Valid test credentials are TBD** — must be provisioned before execution.
 3. **Target application / login URL is TBD** — the ticket does not name the product.
 4. **Exact error-message strings are TBD** — required for deterministic assertion (TC-17).
 5. **SRS document not yet delivered** — SRS-01 to SRS-08 verify a deliverable that does not yet exist.
 6. **Credential rules are TBD** — identifier type (email/mobile/username), case sensitivity (TC-13), min/max length (TC-11/12).
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7. Uncovered Test Cases

None required — all 3 requirement groups are covered by existing test cases.

8. Final Coverage Status

PARTIAL — Coverage Gaps Found

Requirement-level test coverage is 100%, but the ticket lacks acceptance criteria, test data, target environment, and the SRS deliverable, so the overall state is PARTIAL rather than PASS.

9. Human Review Gate

- **What was verified:** 32 test cases (24 login + 8 SRS) map to all 3 KAN-3 requirements extracted from the live JIRA ticket; requirement-level coverage is 100%.
- **What was assumed:** R1–R3 (from the ticket description) are the sole requirements, since the ticket contains no acceptance criteria.
- **What could not be confirmed:** actual execution results (no environment/credentials), and whether SRS-01–08 are valid before an SRS document exists.
- **Which existing tests may need review:** TC-13 (case sensitivity) and TC-19 (session cookie) are marked "TBD" pending spec confirmation.
- **Which new test cases are recommended:** none — coverage is complete at requirement level.

Please review the uncovered/partially covered scenarios and confirm before creating or modifying Jira test cases.